databricks

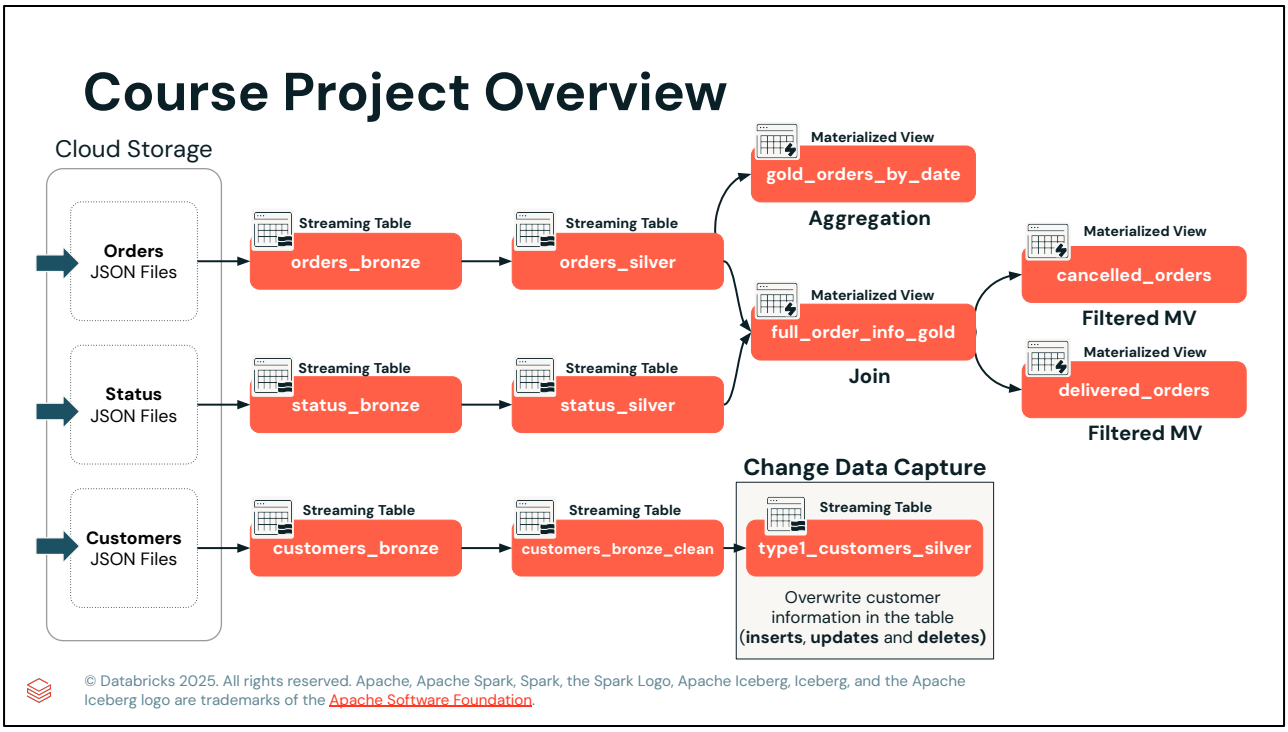
Introduction to Data Engineering in Databricks

LECTURE

Course Project Overview

© Databricks 2025. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

In this lecture, you will get an overview of the course project, where you will build a complete, modular pipeline using Lakeflow Declarative Pipelines to ingest, transform, join, and aggregate order, status, and customer data from JSON files into production-ready tables and views.

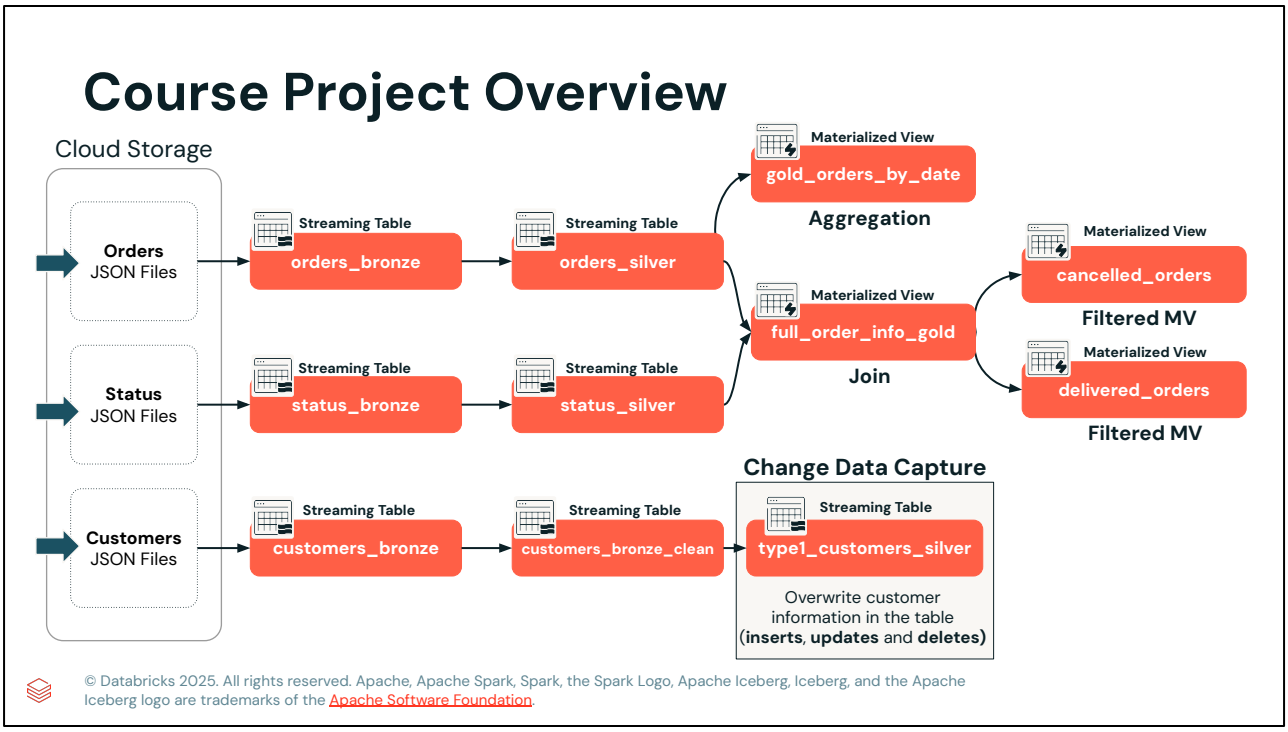


Throughout this course, we'll build a hands-on project together using Lakeflow Spark Declarative Pipelines.

The project begins with files stored in cloud storage. For this course, all files will be in JSON format, but keep in mind that Lakeflow supports many file types in real-world pipelines. We'll build three flows within a single pipeline:

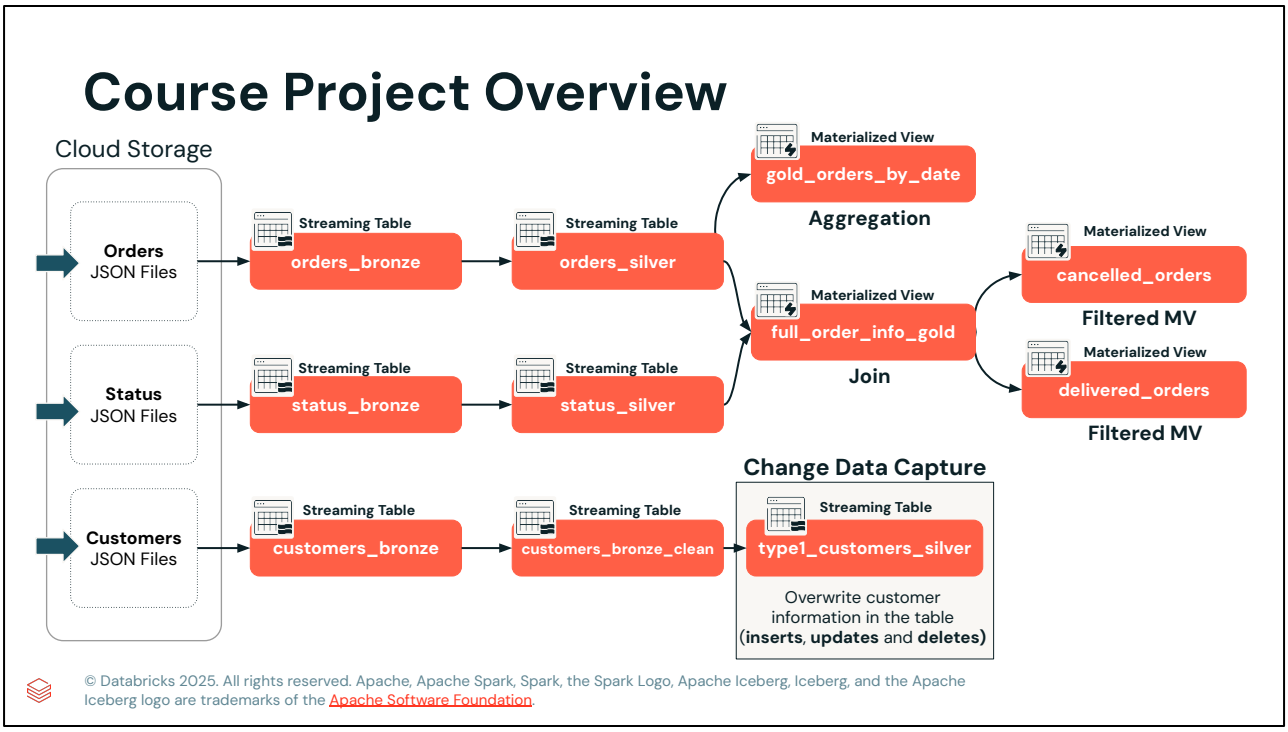
Flow 1: Orders

- Ingests orders JSON files into an **orders_bronze** streaming table.
- Transforms the data into an **orders_silver** streaming table.
- Creates the materialized view **gold_orders_by_date** summarizing the number of orders by date..



Flow 2: Status

- Ingests status JSON files into a **status_bronze** streaming table.
- Transforms the data into a **status_silver** table.
- Joins the orders and status tables to create the materialized view **full_order_info_gold**.
- Produces two materialized views:
  - One for cancelled orders - **cancelled_orders**
  - One for delivered orders - **delivered_orders**



Flow 3: Customer Change Data Capture (CDC)

- Ingests customer JSON files into a **customers_bronze** table.
- Cleans the data into a refined bronze table named **customers_bronze_clean**.
- Performs CDC to track data update changes in the **type1_customers_silver** table.

By the end of the course, you'll have hands-on experience building a complete, modular, and production-ready pipeline architecture using Spark Declarative Pipelines.

© Databricks 2025. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Thank you for completing this lesson and continuing your journey to develop your skills with us.